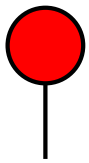
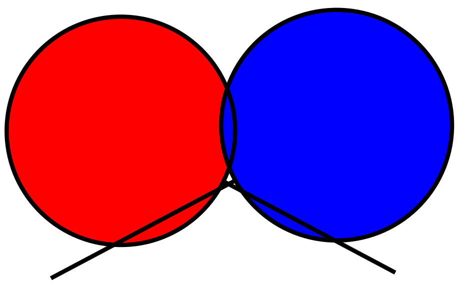
The Lollipop Problem[1]

Neil J. A. Sloane, Dec. 25 2005, revised Dec. 26, 2025



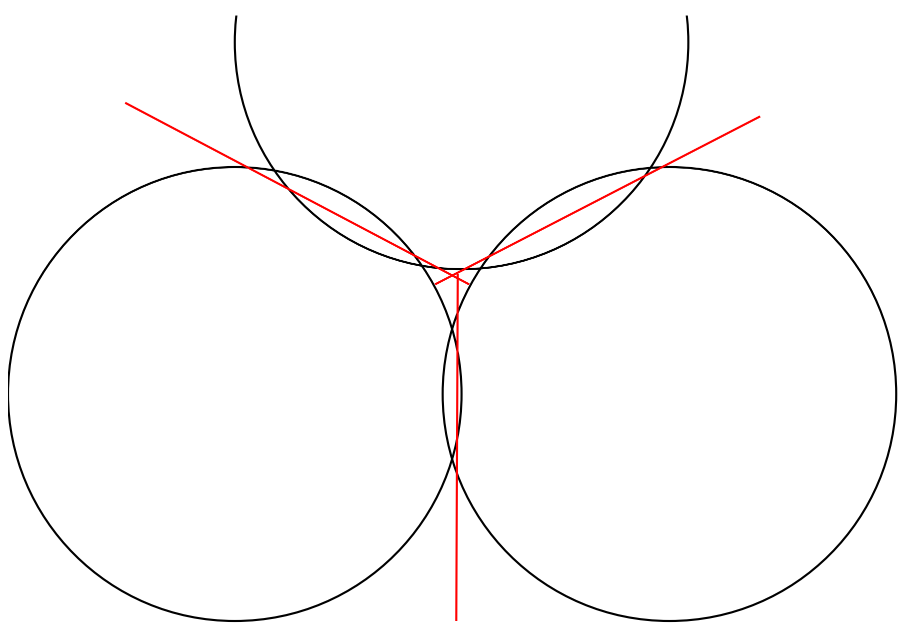
On the left you see a small red lollipop, of the kind that Euclid used to draw in the sand: a circle on a stick. The stick is actually an infinitely long line, or ray, which would start at the center of the circle although that part of the stick that would have been inside the circle is missing (so you don't bite on it).

If you draw this lollipop on a large sheet of paper, you can see that it divides the paper into two regions: inside the red circle, and everything outside the circle. So far, so good: one lollipop, two regions.



What if we have two lollipops? The lollipops can be of any sizes, and the sticks can point in any directions. What is the maximum number of regions they can divide the plane into? It is easy to get 7 regions. But some of us, who have been practicing, can get 10 regions, as shown here (2 red, 2 blue, 1 overlapping lens shape, 3 tiny regions where the sticks cross each other, and 2 infinite regions)! So two lollipops, 10 regions.

With 3 lollipops, the maximum is 25 regions, shown below (some regions are getting pretty small, but that is allowed, and inevitable). So what about 4 lollipops? An upper bound is 46, but I can't even get close. Over to you! Send me your best efforts. SEE NEXT PAGE!



[1] An unsolved problem in Section 11 of *Cutting a pancake with an exotic knife*, by David Cutler and Neil Sloane, https://arxiv.org/abs/2511.15864, presented here as a problem for a family party.

Notes. This is of course a "Find the next term in the sequence" problem. The sequence starts 1 (zero lollipops), 2 (one lollipop), 10, 25, ..., and the following terms are unknown. This is now OEIS entry A389624.

Hints: Counting regions is hard, but we know (see the article in the footnote) that all you have to do is to maximize the total number of crossing points between the different lollipops. If we have n lollipops, numbered 1 through n, let $f(i,j)$ be the number of points where lollipops i and j cross. The total number of crossings is $C = \sum_{i<j} f(i,j)$. Then the number of regions is $R = C + n + 1$. Two lollipops can meet in at most 7 points (see the second picture), so $C \leq 7n(n-1)/2$, $R \leq 7n(n-1)/2 + n + 1$, which is $1, 2, 10, 25, 47, 76, \ldots$. We can get $1, 2, 10, 25$, and it is clear that with 4 lollipops, 47 is impossible.

There were two developments on December 25, 2025. First, at 0:01 am, Hilarie Orman and Rich Schroeppel sent a construction showing that 43 regions are possible with four lollipops, and then at 12:02 pm, Jonas Karlsson sent a construction showing that 44 regions are possible. This is still the record.

With Jonas Karlsson's permission, here are the details of his construction.

The lollipops all have circle radius 1, and their positions are specified by three numbers [x, y, theta], where (x,y) is the exact center of the circle, and theta is the angle in radians of the ray, relative to the x-axis:

[0.95 , 0. , 2.61799388],
[-0.95 , 0. , 0.52359878],
[0.02 , 1.57 , 4.71238898],
[-0.875, 0. , 0.55850536]

Exact values in radians for the four angles are:

theta1 = 5*pi/6
theta2 = pi/6
theta3 = 3*pi/2
theta4 = 8*pi/45

The total number of intersections between the four lollipops is 39, so, by the formula in Eq. (6) of my paper with David Cutler, the number of regions is R = 39 + 4 + 1 = 44. In fact, 44 may be best-possible (the upper bound is 47), but this is not known.

Attached is Jonas's .png plot of the configuration (see the OEIS entry), which can serve as a guide to the positions of the lollipops, although for a proof that the configuration is correct, one must carefully examine the 39 intersection points.

For this purpose, Jonas has provided a text file (also attached to the OEIS entry), giving the 39 intersection points in high precision. A comment that "provenance = [('C1', 'C2')]" means that this point was obtained by intersecting circle 1 and circle 2, etc. Jonas has verified that the resulting planar graph has all the properties that are needed.

I'm also putting this document on my home page, http://neilsloane.com, and I'll try to keep that version up-to-date.